

# Zhuoli Ouyang | Curriculum Vitae

Beijing, China

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## Education

### Southern University of Science and Technology

Undergraduate, Information Engineering

GPA 3.88/4.00 Rank 6/48

- **Fields Elite Class (Honors)**, Department of Mathematics
- **Honors Class**, Department of Electronic & Electrical Engineering

Shenzhen, China

2023 to 2027 (expected)

2023 to 2024 (first year)

2024 to Present

### Dartmouth College

Visiting Student (Research Only)

Advisor Yaoqing Yang

Hanover, NH

Mar 2025 to Oct 2025

### Westlake University

Visiting Student (Research Only)

Advisor Tailin Wu

Hangzhou, China

Oct 2025 to Aug 2026

## Research Experience

Full publications

### NOVA Nullspace Dynamics in Neural Basis Solvers for Moving Boundary PDEs, Transport, Rank Transitions, and Reprojection

Submitted to ICLR 2027

Zhuoli Ouyang (Project Leader), Changxi Chi, Siyuan Li, Tailin Wu

Supervised by Tailin Wu

**Abstract** Enforcing moving boundary constraints is challenging in multiphysics coupling. We propose NOVA to substantially reduce boundary errors while maintaining high interior accuracy. NOVA decomposes neural basis coefficients into a particular solution and a nullspace component. This enforces hard boundary constraints without penalty tuning. To reduce full SVD costs, we derive a horizontal nullspace transport equation. We justify this equation using the Stiefel principal bundle over the Grassmannian. Based on these developments, we design an efficient algorithm. NOVA achieves SOTA boundary accuracy while maintaining high interior accuracy on the evaluated multiphysics benchmarks.

#### My Contributions

- Designed the NOVA framework to enforce hard boundary constraints through nullspace decomposition. Independently completed all theoretical derivations, establishing the mathematical foundation for nullspace transport.
- Designed the transport algorithm based on these derivations to reduce full SVD costs. Independently implemented NOVA, baseline methods, and all experimental code. Organized the complete codebase.
- Conducted the literature review. Designed all benchmarks, baseline comparisons, and ablation studies. Independently ran every experiment. Wrote the full manuscript, covering the method, theoretical analysis, and experimental findings.

### OrthVMS Orthogonal Variational Multiscale Correction for Full-Rank Updates with Low-Rank Optimizer States

Submitted to ICLR 2027

Zhuoli Ouyang (Project Leader), Yujun He, Hongbin Zhong, et al.

Supervised by Zhihai He

**Abstract** LLM training requires substantial GPU memory. Existing low-rank optimizers save memory but underuse gradient information in the orthogonal complement. We propose OrthVMS, a memory-efficient optimizer. Inspired by Variational Multiscale (VMS) in computational mechanics, we treat the projected space as coarse and its orthogonal complement as fine. We construct a full-space preconditioner from decoupled coarse and fine preconditioners. OrthVMS achieves SOTA performance among memory-efficient optimizers on diverse pretraining benchmarks.

#### My Contributions

- Introduced Variational Multiscale (VMS) theory from computational mechanics into LLM optimizer design. Treated the compact subspace as coarse and its orthogonal complement as fine to construct a full-space preconditioner.
- Developed the complete convergence analysis. Laid the theoretical foundation for the paper. Designed the novel algorithm based on the theoretical derivations. Independently implemented OrthVMS, baselines, and the evaluation framework.
- Conducted the literature review. Designed pretraining experiments with progressively increasing sequence lengths. Developed the paper's narrative and structure. Organized the content and wrote the full manuscript.

## Fine-Tuning

Zhuoli Ouyang (Project Leader), Changxi Chi, Siyuan Li, Tailin Wu

Accept (Poster)

Supervised by Tailin Wu

**Abstract** Existing zeroth-order optimization (ZOO) methods for LLM fine-tuning face a trade-off between gradient-estimation variance and curvature information. We propose ENGINE, a memory-efficient ZOO framework for LLM fine-tuning. Inspired by Variational Multiscale (VMS) in computational mechanics, we treat the LoRA space as coarse and the full parameter space as fine. A V-cycle couples the two scales, using full-space curvature feedback to correct low-dimensional updates. We develop a theoretical analysis explaining how ENGINE reduces both variance and bias. ENGINE achieves SOTA performance among ZOO methods on diverse LLM fine-tuning benchmarks, with faster convergence.

## My Contributions

- Formulated ENGINE by extending Variational Multiscale (VMS) theory from computational mechanics to ZOO. Developed the theoretical analysis explaining how coarse/fine coupling reduces both variance and bias.
- Designed the V-cycle algorithm with bidirectional state transfer based on the theoretical derivations. Independently implemented ENGINE, baselines, and the evaluation framework. Organized the complete codebase.
- Conducted the literature review. Designed experiments across model scales and downstream tasks. Developed the paper's narrative and structure. Wrote the full manuscript and handled the entire rebuttal process.

## RMNP Row-Momentum Normalized Preconditioning for Scalable Matrix-Based Optimization [Paper] [Project] [Code]

ICML 2026

Accept (Regular)

Shenyang Deng\*, Zhuoli Ouyang\* (Co-first Author, Lead for Experiments), Tianyu Pang, et al.

\*Equal contribution

Supervised by Yaoqing Yang

**Abstract** RMNP is a matrix optimizer for LLM pretraining that uses row-momentum normalization to compute an  $O(mn)$  preconditioner. It provides efficient matrix updates, with empirical evaluation of preconditioning speed, optimization performance, and training dynamics.

## My Contributions

- Wrote and organized the complete optimizer and benchmark codebase, including training pipelines and baseline implementations for LLM pretraining. Built the experimental setup and independently conducted all experiments.
- Designed benchmarks to evaluate optimization performance, preconditioning costs, and training dynamics across model scales. Analyzed the results to characterize the method's computational efficiency and empirical behavior.
- Conducted the literature review and contributed to parts of the theory and manuscript. Participated throughout the rebuttal process, preparing responses that connected the experimental evidence with the theoretical discussion.

## Honors &amp; Awards

2026 ALT 2026 Best Student Paper Award

2026 ISIT 2026 Strong Accept

2026 DATE 2026 Oral Paper (ACEMARL)

2024 SUSTech, Department of Electronics and Electrical Engineering, Honor Class

2023 SUSTech, Mathematics Department, Fields Elite Class (Top 1%, Honor Class)

## Other Skills

**Programming** Python, C, C++, MATLAB, LaTeX, etc.**Software** PyTorch, PETSc, MATLAB, Fluent**Soccer** Second-class National Athlete in Football**Piano** Level 8, Central Conservatory of Music, Beijing**Hobbies** Exploring mathematics, Coding, Philosophy and History, Playing piano, Playing soccer